Generalization of an Information-Theoretic Measure

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1 Introduction

Spam email classification is a key part in email filtering systems, where the goal is to automatically distinguish between non-spam and spam emails. This task is important in preventing spam from overwhelming inboxes and potentially causing harm to users through phishing, malware, or irrelevant advertisements.

Spam detection systems typically rely on various features extracted from email content, such as the frequency of specific words, the number of capitalized letters, and the presence of special characters. This project investigates the application of information-theoretic measures, Mutual Information [SW49] (MI) and Generalized Mutual Information (GMI), for selecting the most relevant features in a classification task.

The main advantage of using information-theoretic measures in feature selection lies in their ability to quantify the amount of information shared between a feature and the target class. Using the most informative features, we aim to improve the accuracy and efficiency of the classification model.

The aim of this task is to explore the effectiveness of MI and GMI as feature selection methods for spam email classification. Specifically, we will:

1. Calculate MI and GMI for each feature with respect to the target class (spam or non-spam).
2. Select the top features based on MI and GMI.
3. Train a classifier (Logistic Regression) on the selected features and evaluate its performance using accuracy, F1 score, and other relevant metrics.
4. Compare the results obtained with MI and GMI to assess the impact of feature selection on classification performance.

2 Information-Theoretic Measures

2.1 Mutual Information (MI)

MI is a measure from information theory that quantifies the amount of information shared between two random variables. In the context of feature selection, MI is used to evaluate the dependency between a feature and the target class. The higher the MI value, the more information the feature provides about the target class, making it a potentially useful feature for classification tasks.

Mathematically, MI is defined as:

$$MI(X,Y) = \sum_{x \in X} \sum_{y \in Y} p(x,y) log(\frac{p(x,y)}{p(x)p(y)})$$

where:

- $p(x, y)$ is the joint probability distribution of the feature X and target class Y,
- $p(x)$ and $p(y)$ are the marginal probability distributions of X and Y, respectively.

2.2 Generalized Mutual Information (GMI)

Two generalizations of entropy have been proposed by Constantino Tsallis [TMP98] and Alfred Renyi [RV70] using the α parameter to make it more or less sensitive to the shape of probability distributions [MW08]. GMI extends the concept of MI by introducing the parameter α that controls the sensitivity of the measure to rare or infrequent events. This generalization allows for a more flexible measure of the dependence between the feature and the target class.

The formula for GMI is given by:

$$GMI(X, Y, \alpha) = \sum_{x \in X} \sum_{y \in Y} p(x, y)^\alpha log(\frac{p(x, y)^\alpha}{p(x)^\alpha p(y)^\alpha})$$

where:

- $p(x, y)$ is the joint probability distribution of the feature X and target class Y,
- $p(x)$ and $p(y)$ are the marginal probability distributions of X and Y, respectively,
- α is a parameter that adjusts the measure's sensitivity to rare events.

In this project, a default value of $\alpha = 2$ is used, which balances the impact of rare events while still maintaining the core relationship between the feature and target class. GMI allows to explore an alternative to MI that might better capture dependencies, especially when dealing with skewed or imbalanced data.

3 Dataset Description

3.1 Overview

The dataset used in this study is the "SpamBase" dataset, obtained from the UCI Machine Learning Repository [HS99]. This dataset is widely used in the field of machine learning for binary classification tasks, particularly for spam email detection.

The dataset contains 4,601 samples and 57 features, along with a binary target variable indicating whether an email is spam (1) or not (0). Each sample corresponds to an email, and the features are numerical, derived from the content of the emails.

3.2 Features

The features in the dataset can be broadly categorized into the following types:

1. Word frequency features: these are 48 continuous features that represent the percentage of words in the email that match specific keywords (e.g., "make," "address," "all").
2. Character frequency features: these are 6 continuous features representing the percentage of characters in the email that match specific characters (e.g., ;, (, [, !, $, #).
3. Capital letter features:
   - `capital_run_length_average`: average length of uninterrupted sequences of capital letters,
   - `capital_run_length_longest`: length of the longest uninterrupted sequence of capital letters,
   - `capital_run_length_total`: total number of capital letters in the email.

3.3 Target Variable

The target variable, spam, is a nominal variable with two classes:

- 1: the email is classified as spam,
- 0: the email is classified as non-spam.

3.4 Preprocessing

Before applying feature selection and classification, the dataset was preprocessed as follows:

1. Normalization: all features were normalized to bring them to the same scale, ensuring that no single feature dominated the classification process.

2. Data splitting: the dataset was split into training and test sets using an 80-20 ratio. The training set was used to calculate MI and GMI, as well as to train the classifier. The test set was used to evaluate the performance of the model on unseen data.

3.5 Challenges

The dataset presents a slight imbalance in the target variable, with spam emails being slightly less frequent than non-spam emails. This imbalance could influence the performance of the classifier and needs to be considered when interpreting results.

4 Feature Selection Process

The goal of the feature selection process is to identify the most relevant features for spam email classification, reducing dimensionality while retaining the predictive power of the model. This step helps improve model interpretability, reduce overfitting, and improve computational efficiency.

4.1 Methodology

Two information-theoretic measures, MI and GMI, were used to rank the features based on their relevance to the target variable (spam). The top-ranked features were then selected for training the classification model.

Step 1: Calculating MI and GMI

1. MI Calculation: for each feature X_i, MI with respect to the target variable Y was computed using the formula:
$$MI(X_i, Y) = \sum_{x \in X_i} \sum_{y \in Y} p(x,y) log(\frac{p(x,y)}{p(x)p(y)})$$

2. GMI Calculation: for each feature X_i, GMI was calculated using the formula:
$$GMI(X_i, Y, \alpha) = \sum_{x \in X_i} \sum_{y \in Y} p(x,y)^{\alpha} log(\frac{p(x,y)^{\alpha}}{p(x)^{\alpha} p(y)^{\alpha}})$$
The parameter α was set to 2 to balance sensitivity to rare events.

Step 2: Ranking Features
Features were ranked based on their MI and GMI scores in descending order. This ranking reflects the relevance of each feature to the target variable.

Step 3: Selecting Top Features
The top 10 features were selected based on their respective MI and GMI scores for subsequent classification. The choice of the number 10 was motivated by the trade-off between dimensionality reduction and retaining sufficient information for accurate classification.

4.2 Results of Feature Selection

The top 10 features selected by MI and GMI are as follows:

- Top 10 Features by MI and their score:
  1. `word_freq_your (0.1226)`
  2. `word_freq_000 (0.0684)`
  3. `word_freq_you (0.0542)`
  4. `word_freq_remove (0.0460)`
  5. `word_freq_hp (0.0421)`
  6. `word_freq_receive (0.0409)`
  7. `word_freq_all (0.0396)`
  8. `word_freq_our (0.0299)`
  9. `word_freq_hpl (0.0249)`
  10. `word_freq_business (0.0247)`
- Top 10 Features by GMI and their score:
  1. `word_freq_your (0.1952)`
  2. `word_freq_you (0.1293)`
  3. `word_freq_all (0.0490)`
  4. `word_freq_000 (0.0234)`
  5. `word_freq_our (0.0214)`
  6. `word_freq_receive (0.0166)`
  7. `word_freq_hp (0.0139)`
  8. `word_freq_remove (0.0117)`
  9. `word_freq_order (0.0090)`
  10. `word_freq_mail (0.0090)`

5 Classification Experiment

The objective of the classification experiment is to evaluate the effectiveness of the selected features in predicting whether an email is spam or non-spam. Two sets of selected features, one based on MI and the other on GMI, were used to train and test the classification model. Performance metrics were then compared to assess the impact of the feature selection method.

5.1 Methodology

A Logistic Regression classifier was used for this experiment. It was chosen for its simplicity, interpretability, and effectiveness in binary classification problems. This is evaluation setup:

1. Data splitting: the dataset was split into training and test sets in an 80-20 ratio. The training set was used to train the model, while the test set was used to evaluate the model's performance on unseen data.
2. Performance metrics:
   - Accuracy: proportion of correctly classified instances out of the total instances.
   - F1 score: harmonic mean of precision and recall, providing a balanced measure of performance.
   - Confusion matrix: matrix showing true positives, false positives, true negatives, and false negatives.
   - Classification report: includes precision, recall and F1 score for each class.

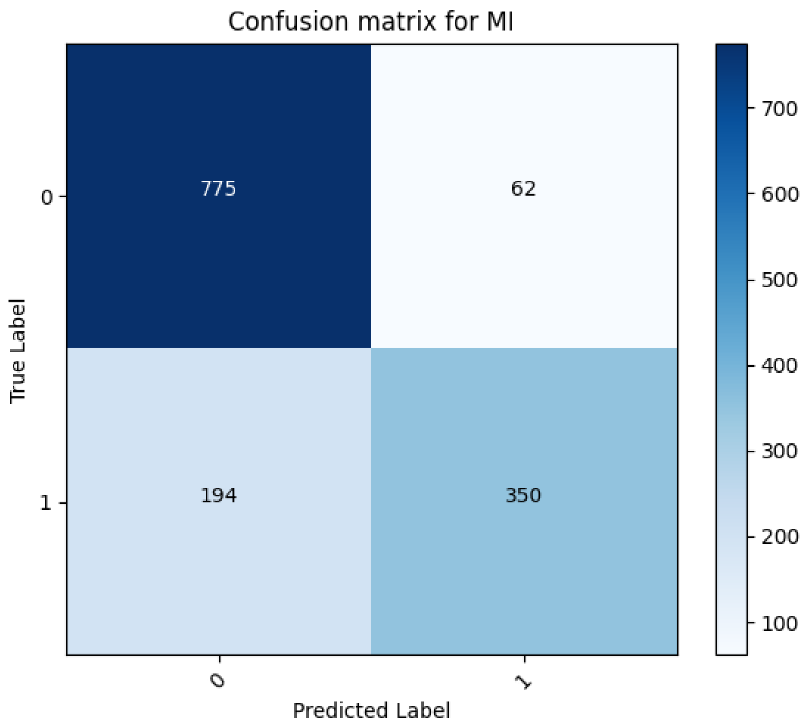

Figure 1: Confusion matrix for MI

5.2 Results

1. MI-based features:

    - Accuracy: 0.8146
    - F1 score: 0.7322
    - Confusion matrix (Figure 1)
    - Classification report (Table 1)

Class	Precision	Recall	F1-Score
0	0.7998	0.9259	0.8583
1	0.8495	0.6434	0.7322

Table 1: Classification report for MI

2. GMI-based features:

    - Accuracy: 0.8074
    - F1 score: 0.7206
    - Confusion matrix (Figure 2)
    - Classification report (Table 2)

Class	Precision	Recall	F1-Score
0	0.7934	0.9223	0.8530
1	0.8407	0.6305	0.7206

Table 2: Classification report for GMI

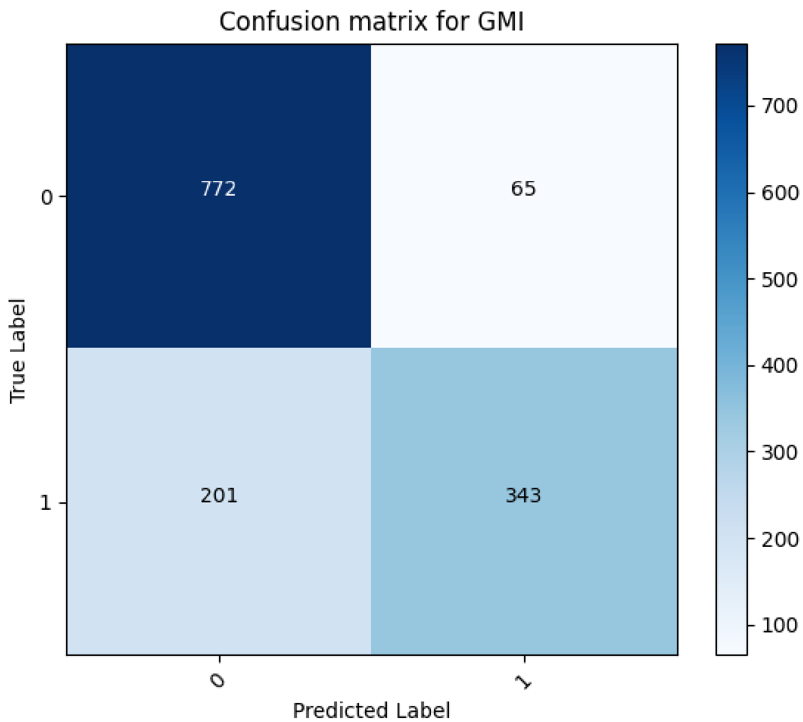

Figure 2: Confusion matrix for GMI

5.3 Analysis

The MI-based feature set achieved higher accuracy (81.46%) compared to the GMI-based feature set (80.74%). The F1 score was also higher for the MI-based feature set (0.7322) compared to the GMI-based feature set (0.7206). This indicates better performance in balancing precision and recall for spam detection. Both feature sets demonstrated similar true negative rates (non-spam emails classified correctly) and true positive rates (spam emails classified correctly), indicating comparable performance in classification accuracy.

The MI-based feature set provided marginally better performance than the GMI-based feature set in this experiment. This suggests that for this dataset, the added flexibility of GMI does not lead to significant improvements in classification performance.

Further experiments on different datasets or with different α values in GMI might reveal scenarios where GMI offers advantages.

6 Conclusion

This study explored the generalization of the Mutual Information (MI) measure, referred to as Generalized Mutual Information (GMI), and its application in feature selection for a spam classification task using the SpamBase dataset.

6.1 Key findings

- Both MI and GMI identified similar sets of top features, with slight variations in their ranking.
- The most informative features for spam detection included frequently occurring words such as 'your,' 'you,' and 'all,' which were prominent in both MI and GMI rankings. Additionally, specific terms like 'remove,' '000,' and 'receive' were identified as significant indicators.
- The MI-based feature set yielded slightly better performance, achieving an accuracy of 81.46% and an F1 score of 0.7322, compared to the GMI-based feature set with an accuracy of 80.74% and an F1 score of 0.7206.

- The MI-based feature set demonstrated better recall for spam detection, which is crucial in minimizing false negatives (missed spam).
- While GMI provides a more flexible framework for feature relevance evaluation by introducing the parameter α, its benefits were not evident in this particular dataset and task.
- GMI's potential might be better realized in datasets with more complex or skewed distributions.

6.2 Future Work

- Apply more complex machine learning models such as Support Vector Machines (SVMs), Random Forests, or Neural Networks to examine how feature selection impacts model performance across various classifiers.
- Test the MI and GMI-based feature selection methods on additional datasets, especially those with imbalanced class distributions or high-dimensional features.
- Explore the impact of different α values in GMI on classification performance.

7 Prospects for Applications of the Generalization to Other Datasets

The GMI is a flexible extension of the traditional MI measure, which can be particularly useful for feature selection in datasets with complex, non-linear relationships between features and the target variable. By adjusting the parameter α, GMI allows for the exploration of different sensitivities to dependencies between variables, making it adaptable for various types of data, including discrete, continuous, or mixed-type features.

In scenarios where datasets present non-linear interactions or where the traditional MI measure may be less sensitive, the ability to fine-tune α in GMI provides an advantage in uncovering more subtle dependencies. Furthermore, this generalization can be applied to domains beyond text classification (e.g., biomedical data, financial forecasting, or sensor networks), where understanding the relationships between features is critical for accurate prediction and feature selection.

As GMI has shown promise in applications like spam classification, its adaptability could be used in other fields like medical diagnosis, fraud detection, and image recognition, particularly in situations where the importance of features may change based on the dataset's characteristics or domain-specific requirements.

8 Software User Manual

8.1 Command-Line Usage

To run the program, use the following command in terminal:

```
python program.py --data_path <data_file_path>
                  --top_features <number_of_features>
                  --alpha <alpha_value>
```

- `--data_path`: path to the input dataset file,
- `--top_features`: the number of top features to select (default is 10),
- `--alpha`: the α value for GMI (default is 2.0).

8.2 Example

Running the program with default settings (top 10 features, α=2):

```
python program.py --data_path spambase.data --top_features 10 --alpha 2
```

References

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